

French Cheese Shop

AI in Requirements Engineering

A French cheese shop exposes how AI systems must interpret vague requests, depend on domain knowledge, and be evaluated under uncertainty.

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Ambiguity
Data
Evaluation



Photo: jackmac34 / Pixabay

Learning Outcomes

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See why a short sentence can hide extra requirements the system still needs.

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Evaluate uncertainty

Judge AI by fit, trust, and usefulness, not only by right or wrong answers.

Customer Query

A Vague Customer Request

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I want something
like Brie,
but stronger.

Easy for a shopkeeper,
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Budget, stock,
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Good answer

Best match, shortlist,
or clear explanation

Conceptual Shift

Traditional RE vs AI-Supported RE

The key shift is from building fixed features to guiding behavior when the system must interpret uncertain input.

Traditional

- Search cheeses by type
- Validate fields and stock
- Same input, same output
- Success = feature works

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- Outputs depend on interpretation
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Requirements are no longer just functions. They also define how the system interprets, learns, and is evaluated.

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
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People infer context.
Software needs it stated.

What does the system still need to know?

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Source: [2] Kiyavitskaya et al. (2008)

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AI systems need these meanings made explicit and testable.

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Challenge 2

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The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

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Data is not just input. It shapes what the system can know and recommend.

Source: [3] Ahmad et al. (2023)

Challenge 3

Challenge 3: Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?

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User input

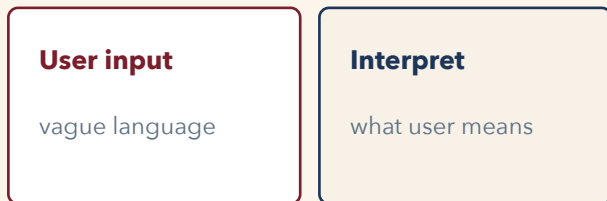
vague language

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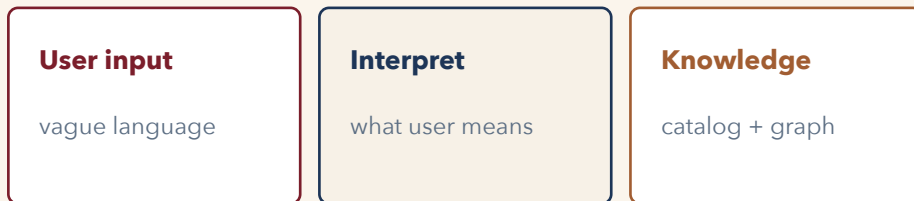


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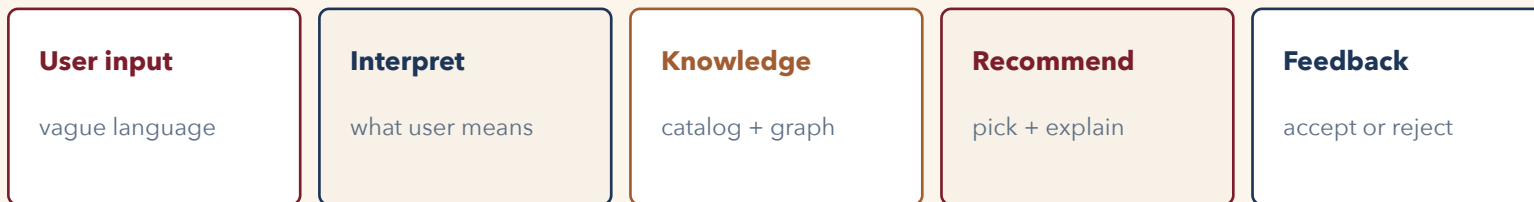


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Main Claim

AI Changes RE

AI changes requirements engineering
from specifying system functions
to shaping behavior, learning, and evaluation.

Questions and Comments

References

Cited sources

[1]

B. Nuseibeh and S. Easterbrook. "Requirements Engineering: A Roadmap." In The Future of Software Engineering, ICSE 2000.

[2]

N. Kiyavitskaya, N. Zeni, L. Mich, and D. M. Berry. "Requirements for tools for ambiguity identification and measurement in natural language requirements specifications." Requirements Engineering, 13(3), 2008.

[3]

K. Ahmad, M. Abdelrazek, C. Arora, M. Bano, and J. Grundy. "Requirements engineering for artificial intelligence systems: A systematic mapping study." Information and Software Technology, 158, 2023.

Architecture Decision Records

Here they work as living design notes that capture key choices for an uncertain system.

Record key assumptions

Write down the system goals, checks, and limits, not only the chosen tool.

Connect goals to code

Link quality goals to prompts, ranking logic, and test criteria.

Keep it testable

Acceptable means it meets today's tests, not that it is always correct.

Example design record

Context: request is vague
Decision: semantic search + reranking
Evaluation: useful and explainable
Checks: stock and domain fit

Good enough for the current shop context.